United States Patent Application Publication	20250270154
Kind Code	A1
Publication Date	August 28, 2025
Inventor(s)	Nauert; Scott Lyle

PROCESS FOR SEPARATING PARAFFINS

Abstract

A process for separating paraffins is disclosed. The process comprises separating a paraffinic stream to provide a first stream comprising C3− hydrocarbons and a second stream comprising C3+ hydrocarbons. The first stream is compressed to provide a first compressed stream. The first compressed stream is fractionated in a fractionation column to provide an overhead stream and a bottom stream comprising C2+ hydrocarbons.

Inventors: **Nauert; Scott Lyle (Chicago, IL)**

Applicant: **UOP LLC** (Des Plaines, IL)

Family ID: **1000008492195**

Appl. No.: **19/050978**

Filed: **February 11, 2025**

Related U.S. Application Data

us-provisional-application US 63557295 20240223

Publication Classification

Int. Cl.: **C07C7/04** (20060101); **C07C7/00** (20060101); **C07C7/09** (20060101)

U.S. Cl.:

CPC **C07C7/04** (20130101); **C07C7/005** (20130101); **C07C7/09** (20130101);

Background/Summary

FIELD

[0001] The field is the separation of light paraffins. The field may particularly relate to separation of paraffins from a naphtha to light paraffinic reactor effluent stream.

BACKGROUND

[0002] Light olefin production is vital to the production of sufficient plastics to meet worldwide demand. Dehydrogenation is a process in which light paraffins such as ethane and propane can be dehydrogenated to make ethylene and propylene, respectively typically in the presence of a catalyst. Dehydrogenation can be achieved in either the presence of an oxidant such as oxygen or in the absence of an oxidant. Non-oxidative dehydrogenation is an endothermic reaction which requires external heat to drive the reaction to completion. Propane dehydrogenation (PDH) is a widely practiced example of non-oxidative dehydrogenation to produce propylene from propane. Ethane oxidative dehydrogenation is a newer oxidative process for converting ethane to ethylene which can be conducted at lower temperatures with lower carbon oxide emissions than non-oxidative and thermal cracking processes.

[0003] Fluid catalytic cracking (FCC) is another endothermic process that can be tuned to produce substantial propylene. However, not every FCC unit is tuned to make substantial propylene. Also, high propylene FCC units do not recover much ethylene; less than 1% of global ethylene supply comes from FCC.

[0004] The great bulk of the ethylene consumed in the production of plastics and petrochemicals such as polyethylene is produced by the thermal cracking of hydrocarbons. Steam is usually mixed with the feed stream to a cracking furnace to reduce the hydrocarbon partial pressure and enhance olefin yield and to reduce the formation and deposition of carbonaceous material in the cracking reactors. The process is therefore often referred to as steam cracking or pyrolysis.

[0005] Paraffins with a range of carbon numbers can be thermally cracked to produce olefins including ethane, propane, butanes, and naphtha. Ethane and naphtha feeds are typical due to higher light olefin yield than propane and butane feeds. Ethane feed is used in regions where light hydrocarbon gases are prevalent. In regions where gas is not abundant, naphtha feed is employed for steam cracking. Naphtha steam cracking has long set the price in the ethylene industry due to higher production cost versus ethane steam cracking. The world does not currently produce enough ethane to supply the growing demand for ethylene. Therefore, regions lacking ethane supply such as Asia and Europe rely mainly on naphtha steam cracking to supply ethylene. Naphtha steam cracking yields only about 30%-35% ethylene with the balance including both relatively high-value by-products comprising propylene, butadiene, and butene-1 and relatively low value by-products comprising pyoil, pygas, and fuel gas. Additional pressures on naphtha steam cracking including minimum production requirements and environmental concerns have led to the withholding of government approvals in certain regions such as China. The ethylene industry needs a more efficient, economical and environmentally friendly route to light olefins from naphtha feeds.

BRIEF SUMMARY

[0006] A process for separating paraffins is disclosed. The process comprises separating a paraffinic stream to provide a first overhead stream comprising C3− hydrocarbons and a second stream comprising C3+ hydrocarbons. The first stream may be compressed to provide a first compressed stream. The first compressed stream is fractionated in a fractionation column to provide an overhead stream and a bottom stream comprising C2+ hydrocarbons. The present process for separating paraffins minimizes the utility duties in the column condensers and the column reboilers for separation.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. **1** is a schematic drawing of an exemplary embodiment of a paraffins separation process and apparatus of the present disclosure.

[0008] FIG. **2** is a schematic drawing of another exemplary embodiment of the paraffins separation process and apparatus of the present disclosure.

[0009] FIG. **3** is a schematic drawing of yet another exemplary embodiment of the paraffins separation process and apparatus of the present disclosure.

DEFINITIONS

[0010] The term "communication" means that fluid flow is operatively permitted between enumerated components, which may be characterized as "fluid communication".

[0011] The term "downstream communication" means that at least a portion of fluid flowing to the subject in downstream communication may operatively flow from the object with which it fluidly communicates.

[0012] The term "upstream communication" means that at least a portion of the fluid flowing from the subject in upstream communication may operatively flow to the object with which it fluidly communicates.

[0013] The term "direct communication" means that fluid flow from the upstream component enters the downstream component without passing through any other intervening vessel.

[0014] The term "indirect communication" means that fluid flow from the upstream component enters the downstream component after passing through an intervening vessel.

[0015] As used herein, the term "predominant" or "predominate" means greater than 50%, suitably greater than 75% and preferably greater than 90%.

[0016] The term "Cx" is to be understood to refer to molecules having the number of carbon atoms represented by the subscript "x". Similarly, the term "Cx−" refers to molecules that contain less than or equal to x and preferably x and less carbon atoms. The term "Cx+" refers to molecules with more than or equal to x and preferably x and more carbon atoms.

[0017] The term "column" means a distillation column or columns for separating one or more components of different volatilities. Unless otherwise indicated, each column includes a condenser on an overhead of the column to condense and reflux a portion of an overhead stream back to the top of the column and a reboiler at a bottom of the column to vaporize and send a portion of a bottoms stream back to the bottom of the column. Feeds to the columns may be preheated. The top pressure is the pressure of the overhead vapor at the vapor outlet of the column. The bottom temperature is the liquid bottom outlet temperature. Overhead lines and bottoms lines refer to the net lines from the column downstream of any reflux or reboil to the column. Stripper columns may omit a reboiler at a bottom of the column and instead provide heating requirements and separation impetus from a fluidized inert media such as steam. Stripping columns typically feed a top tray and take the main product from the bottom.

[0018] As used herein, the term "separator" means a vessel which has an inlet and at least an overhead vapor outlet and a bottoms liquid outlet and may also have an aqueous stream outlet from a boot. A flash drum is a type of separator which may be in downstream communication with a separator that may be operated at higher pressure.

DETAILED DESCRIPTION

[0019] Naphtha and liquefied petroleum gas (LPG) feed stock comprising C3-C8 hydrocarbons is primarily charged to a "Naphtha to Ethane and Propane" (NEP) reactor to convert naphtha and LPG in the presence of hydrogen into desirable ethane and propane along with less desirable methane. Separation of the light paraffins from the reactor effluent may require higher utility duties in the fractionation columns and higher reboiler temperatures which require more expensive utilities. The disclosed process provides a heat integration between the fractionation columns and process streams in the NEP unit and enables using higher-temperature refrigerants for cooling some of the streams.

[0020] Turning to FIG. **1**, an embodiment of a process and an apparatus for separating paraffins **101** is disclosed. The process comprises a naphtha to ethane and propane (NEP) reactor section **11** and a separation unit **111**. The NEP reactor section **11** may comprise a NEP reactor **20** and a splitter column **110**. A naphtha stream in line **12** may be combined with a hydrogen stream in line **14** to provide a charge stream in line **15**. The charge stream in line **15** may be heated and charged to a naphtha to ethane and propane (NEP) reactor **20** to be contacted with an NEP catalyst. In an aspect, a heavy stream may be combined with the naphtha stream in line **12**. The naphtha stream may comprise C4 to C12 hydrocarbons preferably having a T10 between about-10° C. and about 60° C. and a T90 between about 70 and about 180° C. The naphtha feed stream may comprise normal paraffins, iso-paraffins, naphthenes, and aromatics. The naphtha stream may be heated to a reaction temperature of about 300° C. to about 600° C., suitably between about 325° C. and about 550° C., and preferably between about 350° C. and about 525° C. Weight space velocity should be between about 0.3 to about 20 hr.sup.−1, suitably between about 0.5 and about 10 hr.sup.−1 and preferably between about 1 to about 4 hr.sup.−1. A total pressure should be about 0.1 to about 3 MPa (abs), suitably no less than about 1 MPa (abs) and preferably from about 1.5 to about 2.5 MPa (abs). At these conditions, C2-C4 paraffinic yield is consistently in an excess of 80 wt %, while methane yield is less than about 16 wt %, suitably below about 14 wt % and typically below about 12 wt % and preferably no more than 10 wt %. Under these conditions, ethane can make up more than 60 wt % of the total C2 to C3 and for that matter C2 to C4 produced in the NEP reactor **20**.

[0021] The hydrogen-to-hydrocarbon molar ratio is important to producing ethane and propane. The hydrogen-to-hydrocarbon ratio should be about 0.3 to about 15 and preferably about 0.5 to about 5. In a further embodiment, the hydrogen-to-hydrocarbon molar ratio may typically be no more than 5, suitably be no more than 3 and preferably be no more than 2. Low hydrogen-to-hydrocarbon ratio promotes desired reaction kinetics which are initiated with dehydrogenation. Hydrogen-to-hydrocarbon ratio may range from about 50% to about 500%, suitably no more than 300% and preferably no more than 200% of stoichiometric requirements to convert naphtha molecules to ethane and/or propane. The molar ratio of hydrogen to hydrocarbon depends on the feed type including paraffin, olefin, naphthene or aromatics, the feed molecular carbon number, and the desired product between predominantly ethane, predominantly propane or ethane and propane of comparable abundance.

[0022] The NEP catalyst for converting naphtha to ethane and propane may contain a molecular sieve comprising large or medium pore mouths, that is, comprising 10 or 12 member rings, respectively. Examples of suitable molecular sieves include MFI, MEL, MFI/MEL intergrowth, MTW, TUN, UZM-39, IMF, UZM-44, UZM-54, MWW, UZM-37, UZM-8, UZM-8HS. Examples of suitable molecular sieves further include FER, AHT, AEL (SAPO-11), AFO (SAPO-41), MRE, MFS, EUO-1, TON (ZSM-22), MTT (ZSM-23) and UZM-53. Additional molecular sieves with larger pores include FAU, EMT, FAU/EMT intergrowth, UZM-14, MOR, BEA, UZM-50, MTW, ZSM-12. Additional examples include MSE and UZM-35.

[0023] MFI is a suitable NEP catalyst. It will be appreciated that ZSM-5 is an MFI-type aluminosilicate zeolite belonging to the pentasil family of zeolites and having a chemical formula of Na.sub.nAl.sub.nSi.sub.96-nO.sub.192.Math.16H.sub.2O (0<n<10). In various embodiments, the ZSM-5 zeolite may comprise a silica-to-alumina molar ratio of 20 to 1000, 20 to 800, 20 to 600, 20 to 400, 20 to 200 or 20 to 80. In various embodiments, the ZSM-5 zeolite may comprise a crystal size in the range of 10 to 600 nm, 20 to 500 nm, 30 to 450, 40 to 400 nm, or 50 to 300 nm.

[0024] The NEP catalyst may comprise a bound zeolite. The binder may comprise an oxide of aluminum, silicon, zinc, titanium, zirconium and mixtures thereof. The binder may comprise a phosphate in the binder or a phosphate of the forenamed oxide binder materials. Preferably, the binder is a silicon oxide. The MFI zeolite may be supported in a silicon oxide containing binder or an alumina containing binder such as aluminum phosphate.

[0025] MFI zeolite slurry may be first mixed with a binder in the form of colloidal suspension (sol) and gelling reagent and then dropped into hot oil to make spheres controlled to produce 1/888-inch to about 1/32-inch diameter calcined supports. Alternatively, the zeolite may be mixed with a silicon oxide containing binder and extruded to 1/32 to ¼-inch diameter extrudates. Extrudates may be washed with ammonia to remove sodium ions from the zeolite, dried and calcined to remove the organic structural directing agent (OSDA) from the synthesized zeolite. Optionally, the calcined support may be ammonium-ion exchanged using an ammonium nitrate solution to remove residual sodium ions and dried at about 110° C.
[0026] The NEP catalyst comprises a metal on the catalyst. The metal may comprise a transition metal. In a further example, the metal may comprise platinum, palladium, iridium, rhenium, ruthenium and mixtures thereof. The metal may be a noble metal. A modifier metal may also be included on the catalyst. The modifier metal may include tin, germanium, gallium, indium, thallium, zinc, silver and mixtures thereof. The modifier metal should be more concentrated on the binder than on the zeolite. About 0.01 to about 5 wt % of each transition metal and the modifier metal may be on the catalyst.
[0027] Metal may be incorporated into the binder by evaporative impregnation. A solution of platinum such as tetraamine platinate nitrate or chloroplatinic acid may be contacted with the bound spherical or extrudate supports which have been calcined and ion-exchanged in a rotary evaporator, followed by drying and oxidation.
[0028] The NEP catalyst comprises a metal on the bound spherical or extrudate supports of the catalyst. Preferably, more of the metal is on the binder than on the zeolite. At least 60 wt %, suitably at least 70 wt %, preferably at least 80 wt % and most preferably at least 90 wt % of the metal is on the binder. The zeolite and/or the entire NEP catalyst is steam oxidized to drive the metal off the zeolite. Steaming is preferably effected after the metal is added to the catalyst. The dried, impregnated spherical or extrudate supports may be steam oxidized in air for sufficient time to provide NEP catalysts. Steam oxidation in air at a temperature of about 500° C. to about 650° C. and about 5 mol % to about 30 mol % steam for about 1 to 3 hours may be suitable.
[0029] The NEP catalysts must be reduced to activate them for catalyzing the NEP reaction. For example, the catalyst may be reduced in flowing hydrogen at about 500 to about 550° C. for 3 hours before contacting feed.
[0030] After paraffin conversion, a reactor effluent stream comprising paraffins is discharged from the NEP reactor **20** in an effluent line **102**. The reactor effluent stream in line **102** may be a light paraffinic stream. The reactor effluent stream in line **102** may comprise at least about 40 wt % ethane or at least about 40 wt % propane or at least about 70 wt % and preferably at least about 80 wt % ethane and propane. The ethane to propane ratio can range from about 0.1 to about 5. The reactor effluent stream in line **102** can have less than about 16 wt %, suitably less than about 14 wt %, preferably less than about 12 wt %, and more preferably less than about 10 wt % methane.
[0031] The reactor effluent stream in line **102** may be cooled and fed to a splitter column **110** to separate aromatics in a splitter heavy stream. The splitter heavy stream is taken from the splitter column **110** in a bottoms line **114** and may be rich in aromatics. In an embodiment, the splitter heavy stream in line **114** may be recycled to the NEP reactor in charge line **15**.
[0032] A splitter light stream rich in paraffins is produced in an overhead line **116** of the splitter column **110**. A splitter light stream in line **116** may be passed to the separation unit **111** to separate one or more paraffins from the splitter light stream in line **116**. In an embodiment, a splitter side stream may be provided in line **112** from the splitter column **110**. In this embodiment, the splitter side stream in line **112** may comprise single ring aromatics that can be recycled to the NEP reactor in charge line **15**. Moreover, in this embodiment, multi-ring aromatics may be produced in the splitter bottoms line **114**.
[0033] In an exemplary embodiment, the splitter column **110** may comprise two reboilers, an upper reboiler **113** and a lower reboiler **197**. The splitter side stream in line **112** may be collected from a

tray below the feed stage for the reactor effluent stream in line **102**. In an aspect, all the liquid may be collected in the splitter side stream in line **112** from the tray below the feed stage. The splitter side stream in line **112** may be heated in the upper reboiler **113** to provide an upper reboiled stream in line **191**. The upper reboiled stream in line **191** may be recycled back to the splitter column **110** below the draw stage for the splitter side stream in line **112**. In an embodiment, a portion of the upper reboiled stream in line **191** may be taken in line **192** from the splitter column **110**. The remaining portion of the upper reboiled stream in line **191** may be recycled in line **193** back to the splitter column **110** below the draw stage.

[0034] A lower reboiling stream may be taken in lower reboiling line **195** from the splitter heavy stream in line **114**. The lower reboiling stream in line **195** may be heated in the lower reboiler **197** to provide a lower reboiled stream in line **198** which is recyled back to the splitter column **110**. The remaining portion of the splitter heavy stream is taken as net liquid bottoms stream in line **194** from the bottom of the splitter column **110**. In an embodiment, the splitter light stream which may be in the splitter overhead line **116** may be separated into vapor and liquid streams before passing it to the separation unit **111** to separate paraffins. The splitter light stream in line **116** may be cooled in a heat exchanger **117** to provide a cooled splitter light stream in line **118**. The cooled splitter light stream in line **118** is passed to a splitter receiver **119** in which vapor and liquid are separated. A splitter net overhead vapor stream is provided in line **115** from the splitter receiver **119**. The splitter net overhead vapor stream in line **115** is passed to the separation unit **111**. A splitter overhead liquid stream is produced in line **121** from the splitter receiver **119** and passed to the splitter column **110** as a reflux stream.

[0035] The NEP separation unit **111** may be a separator and a fractionation column or a series of fractionation columns and other separation units. The NEP separation unit **111** may comprise a separator **120** and a fractionation column **140**. In an embodiment, the separator **120** may be a fractionation column. In this embodiment, the NEP separation unit **111** may comprise a first fractionation column **120** and a second fractionation column **140**. In an exemplary embodiment, the first fractionation column may be a light ends splitter column **120** which may be a first depropanizer column. In another exemplary embodiment, the second fractionation column **140** may be a demethanizer column **140**.

[0036] The current process comprises a light ends splitter column **120** upstream of the demethanizer column **140** in the separation unit **111**. The upstream light ends splitter column prevents most of the C4+ heavy hydrocarbons from flowing to the demethanizer column **140** and/or a deethanizer column **160**. Otherwise, the presence of C4+ heavy hydrocarbons may lead to higher utility duties in the fractionation columns and higher reboiler temperatures which require more expensive utilities. In addition, C4+ hydrocarbons may have higher viscosity which can reduce heat exchanger performance in the demethanizer column **140** and the deethanizer column **160** and result in more frequent cleaning. In some cases, the presence of significant benzene can lead to freezing in the cold demethanizer column **140**. The current process prevents this by removing C4+ hydrocarbons in the bottoms of the light ends splitter column **120**.

[0037] Referring to the separation unit **111**, the splitter net overhead vapor stream in line **115** is passed to the light ends splitter column **120**. C1-C4 hydrocarbons may be separated in the light ends splitter column **120**. A first overhead stream comprising C3− hydrocarbons may be produced from an overhead in a total overhead line **122** from the light ends splitter column **120**. A second stream comprising C3+ hydrocarbons may be produced from a bottom in a bottoms line **166** from the light ends splitter column **120**. More than about 99 mol % of C2− hydrocarbons may be recovered in the light ends splitter overhead stream of the light ends splitter column overhead line **136**, and more than about 95 mol % of C4+ hydrocarbons may be recovered in the light ends splitter bottoms stream in the light ends splitter bottoms line **166**. The light ends splitter column **120** may be operated at an overhead pressure of about 1034 kPa (gauge) (150 psig) to about 2068 kPa (gauge) (300 psig), and a bottoms temperature of about 80° C. (176° F.) to about 160° C. (320°

F.).

[0038] The first stream comprising C3− hydrocarbons in the light ends splitter total overhead line **122** is heat integrated with a compressor **124**. The first stream in the light ends splitter total overhead line **122** may be compressed in the light ends compressor **124** to provide a first compressed stream in line **126**. In an exemplary embodiment, the first stream in the light ends splitter total overhead line **122** may be compressed to a pressure of about 1724 kPa (gauge) (250 psig) to about 3792 kPa (gauge) (550 psig) in the light ends compressor **124**. In an embodiment, the first stream in the light ends splitter total overhead line **122** is compressed to a pressure of at least about 2.1 MPa (g) (300 psig) or at least about 2.8 MPa (g) (400 psig) or at least about 3.1 MPa (g) (450 psig) to provide a first compressed stream in line **126**. The first compressed stream in line **126** may be passed to a light ends splitter condenser **128** to partially condense the first compressed stream in line **126** and provide a first cooled stream in line **130**. The light ends splitter condenser **128** may comprise one or more heat exchangers. Suitable heat exchange media for condensing the first compressed stream in line **126** in the light ends splitter condenser **128** include air, cooling water, refrigerants such as propane or propylene, and other process streams. The first cooled stream in line **130** may be passed to a light ends splitter receiver **132** for separation into a first light ends net overhead vapor stream in line **136** and a first light ends overhead liquid stream in line **133**. A reflux stream is taken in line **134** from the first light ends overhead liquid stream in line **133** and passed to the light ends splitter column **120**. A net first light ends overhead liquid stream may be taken in line **138** from the first light ends overhead liquid stream in line **133**.

[0039] The overhead compressor **124** for compressing the first stream in the total light ends splitter overhead line **122** serves two benefits: 1) the vapor needs to be compressed to a higher pressure such as about greater than 400 psig for separation in the demethanizer column **140**, and 2) condensing the first stream in the total light ends splitter overhead line **122** at higher pressure increases the temperature at which further separation is conducted. This enables heat integration between the condenser and other process streams in the process and enables use of higher temperature refrigerants which are more economical than the usual lower temperature refrigerants.

[0040] When desired to recover propane as a product stream, more than about 90 mol % of propane may be recovered in the first light ends net overhead vapor stream in line **136** and/or the net first light ends overhead liquid stream in line **138**. When desired to recycle propane to the NEP reactor, then more than about 50 mol % of propane may be recovered in the second stream in the light ends splitter bottoms line **166**, preferably more than about 75 mol % propane may be recovered in the second stream in the light ends splitter bottoms line **166**, and more preferably more than about 90 mol % propane may be recovered in the second stream in the light ends splitter column bottom line **166**.

[0041] The first light ends net overhead vapor stream in line **136** is a high-pressure vapor stream which has a lower concentration of C4+ hydrocarbons. The first light ends net overhead vapor stream in line **136** may be separated in the demethanizer column **140** and optionally a deethanizer column **160** to produce a net gas stream comprising hydrogen and methane, an ethane product stream, and optionally a propane product stream. The second stream in the light ends splitter bottoms line **166** may be optionally separated in another depropanizer column **170** to produce a depropanizer overhead product stream rich in propane in the depropanizer overhead line **172** and a depropanizer bottoms stream rich in C4+ hydrocarbons in depropanizer bottoms line **174**. However, the second depropanizer column may be omitted when propane is recycled to the NEP reactor and in such an embodiment, no depropanized bottoms product stream is produced.

[0042] The first light ends net overhead vapor stream comprising light ends is produced from the receiver **132** in the receiver overhead line **136** and passed to a second fractionation column which is a demethanizer fractionation column **140**. The net first light ends overhead liquid stream in line **138** is also passed to the demethanizer column **140**. The net first light ends overhead liquid stream in line **138** may be passed at a location below the location at which the first light ends net overhead vapor stream in line **136** is passed to the demethanizer fractionation column **140**. The demethanizer fractionation column **140** may be operated at an overhead pressure of about 1241 kPa (gauge) (180 psig) to about 2068 kPa (gauge) (300 psig), and a bottoms temperature of about −30° C. (−86° F.) to about 40° C. (104° F.). The demethanizer fractionation column **140** separates the light hydrocarbons such as methane and hydrogen in a demethanizer overhead stream which is produced in a demethanizer overhead line **142**. A demethanized bottoms stream comprising C2+ hydrocarbons is produced in bottoms line **154** from the demethanizer fractionation column **140**. Optionally, a demethanizer first side stream may be taken in a demethanizer side line **144** from the demethanizer fractionation column **140**. The demethanizer first side stream in line **144** may comprise ethane and propane. The demethanizer first side stream in line **144** may be rich in C2+ paraffins. The demethanizer bottoms stream comprising C2+ hydrocarbons in line **154** may be recovered as a C2 product stream. The demethanizer first side stream in line **144** may be employed in an embodiment that includes a deethanizer column **160**. In an aspect, the demethanizer first side stream in line **144** is a liquid stream. In an embodiment, the demethanizer bottoms stream comprising C2+ paraffins in line **154** and optionally the demethanizer first side stream in line **144** may be further separated to provide an ethane product stream.

[0043] An optional deethanizer column **160** is disclosed for separating ethane from the demethanizer bottoms stream in line **154**. In an aspect, the demethanizer bottoms stream in line **154** may be passed to a demethanizer bottoms exchanger **156** to heat the demethanizer bottoms stream and provide a heated demethanizer bottoms stream in line **158** which is passed to the deethanizer column **160**. The optional demethanizer first side stream in line **144** may also be passed to the deethanizer column **160**. The demethanizer first side stream in line **144** may be passed to a demethanizer sidedraw exchanger **146** to heat the demethanizer first side stream and provide a heated demethanizer first side stream in line **148** which is passed to the deethanizer column **160**. In an aspect, the demethanizer bottoms stream in line **154** and the optional demethanizer first side stream in line **144** may be heat exchanged with the first compressed stream in line **126** in the heat exchanger **128** to provide cooling to the first compressed stream. The heated first side stream in line **148** and the heated demethanizer bottoms stream in line **158** may be taken from the heat exchanger **128** and passed to the deethanizer column **160**. In this embodiment, heat exchangers **128**, **146** and **156** may be one or more heat exchangers in series with the demethanized bottoms stream in line **154** on one side of the heat exchanger and the first compressed stream on the other side of the heat exchanger and optionally the demethanizer first side stream in line **144** on one side of the heat exchanger and the first compressed stream on the other side of the heat exchanger. In an embodiment, heat exchangers **128**, **146**, and **156** may be a single multi-sided heat exchanger with the first compressed stream in line **126** on one side exchanging heat with the demethanized bottoms stream in line **154** on another side and the demethanizer side stream in line **144** on an additional other side. This heat exchanger may be constructed of brazed aluminum, carbon steel, or 300 series stainless steel, and it may be of plate and frame, plate and fin, or spiral wound tube construction. In an embodiment, the heated demethanizer first side stream in line **148** is passed to a side column inlet at a higher location in the deethanizer column **160** than a bottoms column inlet in the deethanizer column for the heated demethanizer bottoms stream in line **158**.

[0044] The deethanizer column **160** produces a deethanizer overhead product stream rich in ethane in the overhead line **162**. An overhead ethane product stream may be provided in the overhead line **162** from the deethanizer column **160**. The deethanizer column **160** may be operated at an overhead pressure of about 1241 kPa (gauge) (180 psig) to about 2068 kPa (gauge) (300 psig), and a bottoms temperature of about 40° C. (104° F.) to about 100° C. (212° F.). A deethanized bottoms stream comprising propane may be taken in a bottoms line **164** from the deethanizer column **160**.

[0045] A deethanizer reboiler stream may be taken in line **165** from the deethanized bottoms stream in line **164**. In an aspect, the splitter light stream in line **116** may be heat exchanged with the deethanized reboiler stream in line **165**. The deethanizer reboiler stream in line **165** may be heated by heat exchange with the splitter light stream in the splitter overhead line **116** in a deethanizer reboiler **167** to provide the cooled splitter light stream in line **118** and a deethanized reboiled stream in line **169**. The heated reboiled stream in line **169** is passed back to the bottom of the deethanizer column **160**. A net deethanized bottoms stream is provided in line **171** rich in propane.

[0046] In an alternate embodiment, the deethanizer reboiler **167** may comprise a stab-in reboiler. In this alternate embodiment, the deethanizer reboiler **167** is incorporated directly in the deethanizer column **160**. In this alternate embodiment, the splitter light stream in the light ends splitter overhead line **116** may be heat exchanged directly with liquid in column **160** in the deethanizer reboiler **167** to provide the cooled light ends splitter stream in line **118**.

[0047] In an aspect, the overhead ethane product stream in line **162** may be charged to an ethylene producing unit **195** in which ethane in the ethane stream is converted into ethylene. In an embodiment, the ethylene producing unit **195** may include a steam cracking unit. The overhead ethane product stream in line **162** may be cracked under steam in a furnace to produce a cracked stream including an ethylene stream in line **196**. The overhead ethane product stream in line **162** may be charged to the ethane steam cracking unit in the gas phase. The ethane steam cracking unit **195** may preferably be operated at a temperature of about 750° C. (1382° F.) to about 950° C. (1742° F.). The cracked stream exiting the furnace of the ethane steam cracking unit **195** may be in a superheated state. One or more quench columns, or other devices but preferably an oil quench column and/or a water quench column, may be used for quenching or separating the cracked stream into a plurality of cracked streams. The ethane steam cracking unit **195** may further comprise additional distillation columns, amine wash columns, compressors, expanders, etc. to separate the cracked stream into cracked streams rich in individual light olefins, the most predominant of which is the ethylene stream in line **196**. The ethylene stream in line **196** may comprise a yield of at least 75 wt %, preferably at least 80 wt %, ethylene based on the overhead ethane product stream in line **162**. Among the other components in the cracked stream exiting the ethane steam cracking unit **195** may be hydrogen, methane, propylene, butene, and pyrolysis gas. Each of these components may be recovered and further processed.

[0048] The ethylene stream in line **196** and perhaps a propylene stream from the ethylene producing unit may be recovered or transported to polymerization plants, chemical plants or exported. A butene stream may be recovered and used to produce plastics or other petrochemicals by processes such as polymerization or exported. Product recovery of at least 50 wt %, typically at least 60 wt % and suitably at least 70 wt % of valuable ethylene, propylene, and butylene products is achievable from the ethane steam cracking unit **195** based on the overhead ethane product stream in line **162**.

[0049] The present disclosure provides an optional depropanizer column **170** for separating a propane product stream from the light ends bottoms stream in line **166**. The depropanizer column **170** may be deemed a second depropanizer column if the light ends splitter column **120** is deemed a first depropanizer column. A depropanizer overhead stream comprising propane is separated in an overhead line **172** in the depropanizer column **170**. A depropanized bottoms stream rich in C4+ paraffins is separated in a depropanizer bottoms line **174** from the depropanizer column **170**. The depropanized overhead stream in line **172** may be combined with the net deethanized bottoms stream in line **171** to provide a propane product stream in line **182**.

[0050] In an aspect, the propane product stream in line **182** may be charged to a propylene producing unit **185** in which propane in the propane product stream is converted into propylene. The propylene producing unit **185** may be a propane dehydrogenation (PDH) unit. PDH catalyst is used in a dehydrogenation reaction process to catalyze the dehydrogenation of propane. The conditions in the dehydrogenation reactor may include a temperature of about 500 to about 800° C., a pressure of about 40 to about 310 kPa (abs) and a catalyst to oil ratio of about 5 to about 100.

[0051] The dehydrogenation reaction may be conducted in a fluidized manner such that gas, which may comprise the reactant paraffins with or without a fluidizing inert gas, is distributed to the reactor in a way that lifts the dehydrogenation catalyst in the reactor vessel while catalyzing the dehydrogenation of paraffins. During the catalytic dehydrogenation reaction, coke is deposited on the dehydrogenation catalyst leading to reduction of the activity of the catalyst. The dehydrogenation catalyst must then be regenerated in a regenerator. The regenerator may combust coke from the dehydrogenation catalyst and fuel gas to ensure sufficient enthalpy in the dehydrogenation reactor to promote the endothermic reaction.

[0052] The dehydrogenation catalyst selected should minimize cracking reactions and favor dehydrogenation reactions. Suitable catalysts for use herein include an active metal which may be dispersed in a porous inorganic carrier material such as silica, alumina, silica alumina, zirconia, or clay. An exemplary embodiment of a catalyst includes alumina or silica-alumina containing gallium, a noble metal, and an alkali or alkaline earth metal.

[0053] The catalyst support comprises a carrier material, a binder and an optional filler material to provide physical strength and integrity. The carrier material may include alumina or silica-alumina. Silica sol or alumina sol may be used as the binder. The alumina or silica-alumina generally contains alumina of gamma, theta and/or delta phases. The catalyst support particles may have a nominal diameter of about 400 to about 5000 micrometers with the average diameter of about 600 to about 3500 micrometers. Preferably, the surface area of the catalyst support is about 85 to about 140 m2/g.

[0054] The fluidized dehydrogenation catalyst may comprise a dehydrogenation metal on a support. The dehydrogenation metal may be a one or a combination of transition metals. A noble metal may be a preferred dehydrogenation metal such as platinum or palladium. Gallium is an effective metal for paraffin dehydrogenation. Metals may be deposited on the catalyst support by impregnation or other suitable methods or included in the carrier material or binder during catalyst preparation.

[0055] The acid function of the catalyst should be minimized to prevent cracking and favor dehydrogenation. Alkali metals and alkaline earth metals may also be included in the catalyst to attenuate the acidity of the catalyst. Rare earth metals may be included in the catalyst to control the activity of the catalyst. Concentrations of 0.001% to 10 wt % metals may be incorporated into the dehydrogenation catalyst. In the case of the noble metals, it is preferred to use about 10 parts per million (ppm) by weight to about 600 ppm by weight noble metal. More preferably it is preferred to use about 10 to about 100 ppm by weight noble metal. The preferred noble metal is platinum. Gallium should be present in the range of 0.3 wt % to about 3 wt %, preferably about 0.5 wt % to about 2 wt %. Alkali and alkaline earth metals may be present in the range of about 0.05 wt % to about 1 wt %.

[0056] Regenerated catalyst may be contacted with the propane stream perhaps with a fluidizing gas to lift the propane stream and dehydrogenation catalyst up a riser while dehydrogenation occurs. Above the riser spent dehydrogenation catalyst and propylene product may be separated by a centripetal separation device. Propylene product gas may be quenched with a cooling fluid to prevent over reaction to undesired by-products. Separation of the propylene product from the PDH effluent stream may include quench contacting and fractionation to produce a propylene product stream in line **186**. Unreacted propane may be recycled to the dehydrogenation reactor and lighter gases may be recycled to the regenerator as fuel gas to be combusted to provide enthalpy for the reaction.

[0057] The propylene producing unit **185** may also employ a catalytic moving bed reactor. The reactor section may comprise several radial flow reactors in parallel or series heated by charge and interstage heaters. The propane stream perhaps with added hydrogen flows in each dehydrogenation reactor from a screened center pipe through an annular dehydrogenation catalyst bed to an outer effluent annulus. Flow may be in the reverse fashion. The dehydrogenation catalyst may comprise a noble metal or mixtures thereof, a modifier selected from the group consisting of alkali metals or alkaline-earth metals and mixtures thereof, a component selected from the group consisting of tin, germanium, lead, indium, gallium, thallium, and mixtures thereof, and a porous support forming a catalyst particle. The catalyst support may comprise oil dropped alumina spheres.

[0058] Dehydrogenation conditions may include a temperature of from about 400 to about 900° C., a pressure of from about 0.01 to 10 atmospheres absolute, and a liquid hourly space velocity (LHSV) of from about 0.1 to 100 hr-1. The pressure in the dehydrogenation reactor is maintained as low as practicable, consistent with equipment limitations, to maximize chemical equilibrium advantages. Spent dehydrogenation catalyst in the annular catalyst bed may be withdrawn from the bottom of the bed, forwarded to a regenerator to combust coke from the catalyst with air at about 450 to about 600° C. Noble metal on the catalyst may be redispersed by an oxyhalogenation process, dried and returned to the top of the dehydrogenation catalyst bed as regenerated dehydrogenation catalyst.

[0059] Dehydrogenation effluent from the propylene producing unit **185** may be cooled, compressed, dried and hydrogen is cryogenically separated from the hydrocarbons with a net gas purity of 85 to 93 mol % hydrogen. Hydrocarbon liquid is selectively hydrogenated to convert diolefins and acetylenes and the hydrocarbon liquid is fractionated in a deethanizer column to remove ethane and propylene is split from propane in a propane-propylene splitter column to provide polymer-grade propylene in line **186**. Propane may be recycled as feed to the propylene producing unit **185**. FIG. **2** illustrates another embodiment **201** of a process and an apparatus for separating paraffins. In the embodiment as shown in FIG. **2**, the first stream comprising C3− hydrocarbons is cooled and separated before compressing it in the compressor **124** of the separation unit **211**. Elements in FIG. **2** with the same configuration as in FIG. **1** will have the same reference numeral as in FIG. **1**. The configuration and operation of the embodiment of FIG. **2** is essentially the same as in FIG. **1** with the following exceptions.

[0060] As shown in FIG. **2**, the splitter net overhead vapor stream in line **115** from FIG. **1** may be passed to the light ends splitter column **120**. C1-C4 hydrocarbons may be separated in the light ends splitter column **120**. A first overhead stream comprising C3− hydrocarbons may be taken from an overhead in a total overhead line **122** from the light ends splitter column **120**. A second stream comprising C3+ hydrocarbons may be taken from a bottom in a bottoms line **166** from the light ends splitter column **120**. More than about 99 mol % of C2− hydrocarbons may be recovered in the light ends splitter overhead stream of the light ends splitter column overhead line **136**, and more than about 95 mol % of C4+ hydrocarbons may be recovered in the light ends splitter bottoms stream in the light ends splitter bottoms line **166**.

[0061] The first overhead stream in line **122** may be passed to a light ends splitter condenser **128** to cool and partially condense the first stream in line **122**. A cooled first stream may be provided in line **230** from the light ends splitter condenser **128**. The light ends splitter condenser **128** may comprise one or more heat exchangers. Suitable heat exchange media for condensing the first overhead stream in line **122** in the light ends splitter condenser **128** include air, cooling water, refrigerants such as propane or propylene, and other process streams. The cooled first stream in line **230** may be passed to the light ends splitter receiver **132** for separation into a second light ends net overhead vapor stream in line **236** and a second light ends overhead liquid stream in line **234**. In an aspect, an entirety of the second light ends overhead liquid stream in line **234** may be passed to the light ends splitter column **120**.

[0062] The second light ends net overhead vapor stream in line **236** may be compressed in the light ends compressor **124** to provide a second compressed stream in line **238**. In an embodiment, the second light ends net overhead vapor stream in line **236** is compressed to a pressure of at least about 2.1 MPa (g) (300 psig) or at least about 2.8 MPa (g) (400 psig) or at least about 3.1 MPa (g) (450 psig) to provide a second compressed stream in line **238**. The second compressed stream in line **238** may be passed to the demethanizer column **140** as described for FIG. **1**. The rest of the process is same as previously described in FIG. **1**.

[0063] Another embodiment **301** of a process and an apparatus for separating paraffins is illustrated in FIG. **3**. In the embodiment as shown in FIG. **3**, the separation unit **311** comprises a first separator **320** and a second separator **332**. Elements in FIG. **3** with the same configuration as in FIG. **1** will have the same reference numeral as in FIG. **1**. The configuration and operation of the embodiment of FIG. **3** is essentially the same as in FIG. **1** with the following exceptions.

[0064] A splitter net overhead vapor stream is produced in line **115** from the splitter receiver as shown in FIG. **1**. The splitter net overhead vapor stream in line **115** is passed the separation unit **311**. As described later in detail, the splitter net overhead vapor stream in line **115** may be combined with a recycle liquid stream in line **334** to provide a combined stream in line **316** which is separated in the separation unit **311**.

[0065] In the embodiment as shown in FIG. **3**, the combined stream in line **316** may be passed to a first separator **320**. In the first separator **320**, C3− hydrocarbons are separated in a first separated vapor stream which may be produced in the overhead in line **322**. C3+ hydrocarbons may be produced in a first bottom liquid stream in a bottoms line **366** from the first separator **320**. The first separator **320** may be operated at a pressure of about 1034 kPa (gauge) (150 psig) to about 1724 kPa (gauge) (250 psig), and a temperature of about 20° C. (68° F.) to about 50° C. (122° F.). In an exemplary embodiment, the first separator **320** may be a first knock out drum (KOD).

[0066] The first separated vapor stream in line **322** may be compressed in the light ends compressor **124** to provide a first compressed stream in line **326**. In an exemplary embodiment, the first separated stream in line **322** may be compressed to a pressure of about 1724 kPa (gauge) (250 psig) to about 3792 kPa (gauge) (550 psig) in the light ends compressor **124**. In an embodiment, the first separated vapor stream in line **322** is compressed to a pressure of at least about 2.1 MPa (g) (300 psig) or at least about 2.8 MPa (g) (400 psig) or at least about 3.1 MPa (g) (450 psig) to provide a first compressed stream in line **326**.

[0067] The first compressed stream in line **326** may be passed to the light ends splitter condenser **128** to partially condense the first compressed stream in line **326** and provide a first cooled stream in line **330**. The light ends splitter condenser **128** may comprise one or more heat exchangers. Suitable heat exchange media for condensing the first compressed stream in line **326** in the light ends splitter condenser **128** include air, cooling water, refrigerants such as propane or propylene, and other process streams. In an exemplary embodiment, the first compressed stream in line **326** may be cooled to a temperature of about 30° C. (86° F.) to about 50° C. (122° F.) in the light ends splitter condenser **128**. The first cooled stream in line **330** may be passed to a second separator **332** for separating the light ends such as C3− hydrocarbons.

[0068] In second separator **332**, the first cooled stream in line **330** is separated into a second separated vapor stream comprising C3− hydrocarbons in line **336**. The second bottom liquid stream in line **334** is taken from the bottom of the second separator **332** and recycled to the first separator **320**. The second bottom liquid stream in line **334** may be combined with the splitter net overhead vapor stream in line **115** and recycled to the first KOD **320** as earlier described. The second separator **332** may be operated at a pressure of about 1724 kPa (gauge) (250 psig) to about 3792 kPa (gauge) (550 psig), and a temperature of about 30° C. (86° F.) to about 50° C. (122° F.). In an exemplary embodiment, the second separator **332** may be a second KOD.

[0069] The second separated vapor stream in line **336** is passed to the second fractionation column **140** which may be a demethanizer fractionation column **140**. The demethanizer fractionation column **140** may be operated at an overhead pressure of about 1241 kPa (gauge) (180 psig) to about 2068 kPa (gauge) (300 psig), and a bottoms temperature of about −30° C. (−22° F.) to about 40° C. (104° F.). The demethanizer fractionation column **140** separates the light hydrocarbons such as methane and hydrogen in a demethanizer overhead stream which is produced in a demethanizer overhead stream in line **142** from the demethanizer fractionation column **140**. A demethanized bottoms stream comprising C2+ hydrocarbons is produced in a bottoms line **154** from the demethanizer fractionation column **140**. Optionally, a first side stream may be taken in a demethanizer side line **144** from the demethanizer fractionation column **140**. The first side stream in the demethanizer side line **144** and the demethanized bottoms stream in the bottoms line **154** may be cooled and separated in the deethanizer column **160** of FIG. **1**. The rest of the process is the same as previously described in FIG. **1**.

EXAMPLES

Example 1

[0070] The splitter column **110** of FIG. **1** was simulated using process modeling software. Table 1 provides simulated stream properties and compositions for the light paraffinic stream in line **102**.

TABLE-US-00001 TABLE 1 Stream 102 Temperature (° C.) 130 Pressure (kPa(a)) 1712 Vapor Fraction 0.956 mass flow (kg/h) 100,000 Composition (wt %) Hydrogen 1.0% Methane 5.0% Ethane 27.0% Propane 27.0% n-Butane 5.0% Toluene 32.0% Naphthalene 3.0%

[0071] The reactor effluent stream in line **102** was fed to a trayed distillation column **110**. The splitter light stream in line **116** was cooled to about 40° C. in the splitter condenser **117** then passed to the splitter receiver **119** to separate the splitter net overhead vapor stream comprising C4− in line **115** and the splitter overhead liquid stream in line **121** which was returned to the top of the column. All column liquid was collected below the feed stage, heated to 197° C. in an upper reboiler **113**, and returned to the column at the tray below the draw stage. The splitter side stream was drawn from the upper reboiler return tray. A portion of the column bottoms in line **195** were reboiled in a lower reboiler **197** and returned to the column in a lower reboiled stream in line **198**. The net liquid bottoms stream is taken as the splitter heavy stream in line **194**. The splitter condenser **117** duty for this separation was −21.3 GJ/h which may be provided by cooling water or another suitable cooling medium. The upper reboiler duty was 5.2 GJ/h, and the lower reboiler duty was 1.1 GJ/h. The splitter net overhead vapor stream in line **115** is a suitable feed to the separation unit described in following examples.

Example 2

[0072] The paraffins separation process **311** illustrated in FIG. **3** was simulated using process modeling software. The splitter net overhead vapor stream of Example 1 was used as the light paraffins stream in line **115** which was fed to the process **301**. The light paraffins stream in line **115** was combined with a recycle liquid stream in line **334** and passed to a compressor suction separator **320** to separate a first vapor product stream in line **322** and a liquid product stream in line **366** comprising predominantly C3+ hydrocarbons. The first vapor product in line **322** was passed to a compressor **124** and compressed to 3503 kPa (gauge) (508 psig) to provide a compressed vapor product stream in line **326**. The compressed vapor product stream was passed to a compressor discharge cooler **328** and cooled to 40° C. to provide a cooled compressed vapor product stream in line **330**. The cooled compressed vapor product stream was passed to a compressor discharge separator **332** to provide a second vapor product stream in line **336** comprising predominately C3− hydrocarbons and the recycle liquid stream in line **334**. The second vapor product stream is passed to a demethanizer column **140** to provide a demethanizer overhead stream in line **142**, a demethanizer side stream in line **144**, and a demethanizer bottoms stream in line **154**. The compressor **124** process work was 1,860 kW to compress the first vapor product to 508 psig. The second vapor product stream in line **336** processed in the demethanizer column had a significant amount of C4+ hydrocarbons of about 7.5 wt % which increases the stream viscosity and demethanizer reboiler temperature.

Example 3

[0073] The paraffins separation process **201** illustrated in FIG. **2** was simulated using process modeling software. Table 3 provides simulated stream properties and compositions for select streams. The splitter net overhead vapor stream of Example 1 was used as the light paraffins stream in line **115** which was passed to the light ends splitter column **120**. The light ends splitter total overhead stream in line **122** was condensed in condenser **128** at −4° C. and separated into liquid reflux second light ends overhead liquid stream in line **234** and second light ends net overhead vapor stream in line **236** in light ends splitter receiver **132**. The second light ends net overhead vapor stream in line **236** was compressed to 3447 kPa (gauge) (500 psig) in light ends compressor **124** and passed to demethanizer column **140** to produce the demethanizer overhead stream in line **142**, the demethanizer side stream in line **144**, and the demethanizer bottoms stream in line **154**. The light ends splitter condenser **128** duty was −10.1 GJ/h. The light ends compressor **124** process work is 1,380 kW to compress the second light ends net overhead vapor stream in line **236** to 500 psig.

[0074] Compared with Example 2, less than 5% of the C4 hydrocarbons and no C5+ hydrocarbons were passed to the demethanizer column to be fractionated resulting in 13% less feed to the demethanizer column by mass. This was due to better separation efficiency in the light ends splitter column **120** than can be achieved with a series of separator vessels as in Example 2. As a result, the demethanizer reboiler temperature decreased by 5° C. accompanied by a reduction in reboiler duty. The reduced feed rate also decreased the size of the demethanizer column and associated heat exchange equipment, and it reduces the refrigerant duty required to cool the feed and provide condenser duty to drive the separation. Another benefit of the paraffins separation process **201** in Example 3 is that the compressor work required to compress the demethanizer feed is 26% lower than in Example 2. The demethanizer side stream in line **144** may be heat exchanged with the light ends splitter total overhead stream to provide condensing duty for the light ends splitter condenser **128**. Heating the demethanizer side stream to −8° C. to provide sufficient approach temperature for heat exchange provides 0.5 GJ/h duty which is 5% of the total condensing duty required in light ends splitter condenser **128**.

Example 4

[0075] The paraffins separation process and apparatus **101** illustrated in FIG. **1** was simulated using process modeling software. In the simulation, the splitter net overhead vapor stream in line **115** was passed to the light ends splitter column **120**. The first stream in the light ends splitter total overhead line **122** was compressed in light ends compressor **124**, condensed in light ends splitter condenser **128** at 23° C., and separated into liquid reflux stream in line **134** and the first light ends net overhead vapor stream in line **136** in light ends splitter receiver **132**. The first light ends net overhead vapor stream in line **136** was passed to the demethanizer column **140** to produce the demethanizer overhead stream in line **142**, the demethanizer side stream in line **144**, and the demethanizer bottoms stream in line **154**. The light ends splitter condenser **128** duty was −16.5 GJ/h. The light ends compressor **124** process work was 2,075 kW to compress the light ends total overhead vapor stream to 500 psig.

[0076] The paraffins separation process and apparatus **101** of Example 4 demonstrates similar benefits over Example 2 with respect to the demethanizer column. The light ends splitter net overhead stream fed to the demethanizer column was 13% smaller by mass with less than 5% of the C4 hydrocarbons and no C5+ hydrocarbons. The reduced feed rate and significantly less C4+ hydrocarbons resulted in a smaller demethanizer column and heat exchange equipment, lower reboiler temperature, lower reboiler duty, and less refrigerant duty required to cool the demethanizer feed stream and provide condenser duty to drive the separation. A difference from Example 3 is that the separations process and apparatus **101** compresses the light ends splitter total overhead stream rather than the light ends splitter net overhead stream. The light ends compressor process work for separation process and apparatus **101** of Example 4 was 50% higher than for separation process **201** of Example 3 because more gas was compressed, but a benefit is that the light ends splitter condenser **128** was operated at a higher temperature of 23° C. compared to −4° C. in Example 3. This allowed for higher temperature utilities to be used for condensing duty, or it can enable better heat integration with other process streams such as the demethanizer side stream in line **144**. The demethanizer side stream in line **144** may be heat exchanged with the light ends splitter total overhead stream to provide condensing duty for the light ends splitter condenser **128**. Heating the demethanizer side stream to 15° C. to provide sufficient approach temperature for heat exchange provided 7.0 GJ/h duty which is 42% of the total condensing duty required in light ends splitter condenser **128**.

Specific Embodiments

[0077] While the following is described in conjunction with specific embodiments, it will be understood that this description is intended to illustrate and not limit the scope of the preceding description and the appended claims.

[0078] A first embodiment of the present disclosure is a process for separating paraffins comprising separating a paraffinic stream to provide a first stream comprising C3− hydrocarbons and a second stream comprising C3+ hydrocarbons; compressing the first stream to provide a first compressed stream; and fractionating the first compressed stream in a fractionation column to provide an overhead stream and a bottom stream comprising C2+ hydrocarbons. An embodiment of the present disclosure is one, any or all of prior embodiments in this paragraph up through the first embodiment in this paragraph further comprising cooling the first compressed stream to provide a first cooled stream; separating the first cooled stream to provide a first vapor stream and a first liquid stream; and passing the first vapor stream to the fractionation column. An embodiment of the present disclosure is one, any or all of prior embodiments in this paragraph up through the first embodiment in this paragraph further comprising passing a portion of the first c liquid stream to the fractionation column. An embodiment of the present disclosure is one, any or all of prior embodiments in this paragraph up through the first embodiment in this paragraph further comprising passing the second stream to a depropanizer column to provide an overhead stream comprising propane and a depropanized bottom stream comprising C4+ hydrocarbons. An embodiment of the present disclosure is one, any or all of prior embodiments in this paragraph up through the first embodiment in this paragraph further comprising passing the bottom stream comprising C2+ hydrocarbons to a deethanizer column to provide an overhead product stream comprising ethane, and a deethanized bottoms stream comprising propane; and combining the deethanized bottoms stream with the overhead stream comprising propane to provide a propane product stream. An embodiment of the present disclosure is one, any or all of prior embodiments in this paragraph up through the first embodiment in this paragraph further comprising fractionating a reactor effluent stream in a splitter column to separate aromatics in a splitter heavy stream and provide a splitter light stream comprising paraffins; and taking the paraffinic stream from the splitter light stream. An embodiment of the present disclosure is one, any or all of prior embodiments in this paragraph up through the first embodiment in this paragraph further comprising contacting a naphtha stream with a catalyst in the presence of hydrogen to produce paraffins; and taking the reactor effluent stream from the paraffins. An embodiment of the present disclosure is one, any or all of prior embodiments in this paragraph up through the first embodiment in this paragraph further comprising recycling the splitter heavy stream to the contacting step. An embodiment of the present disclosure is one, any or all of prior embodiments in this paragraph up through the first embodiment in this paragraph further comprising heat exchanging the first compressed stream with the bottom stream comprising C2+ hydrocarbons to cool the first compressed stream and heat the bottom stream comprising C2+ hydrocarbons. An embodiment of the present disclosure is one, any or all of prior embodiments in this paragraph up through the first embodiment in this paragraph further comprising taking a first side stream from the fractionation column; and fractionating the first side stream in the deethanizer column. An embodiment of the present disclosure is one, any or all of prior embodiments in this paragraph up through the first embodiment in this paragraph further comprising heat exchanging the first compressed stream with the first side stream to cool the first compressed stream and heat the first side stream. An embodiment of the present disclosure is one, any or all of prior embodiments in this paragraph up through the first embodiment in this paragraph further comprising heat exchanging the first compressed stream with the bottom stream comprising C2+ hydrocarbons and the first side stream to cool the first compressed stream and heat the bottom stream comprising C2+

hydrocarbons and the first side stream. An embodiment of the present disclosure is one, any or all of prior embodiments in this paragraph up through the first embodiment in this paragraph further comprising heat exchanging the splitter light stream with a deethanized bottoms stream in a deethanizer reboiler to cool the splitter light stream and provide a cooled splitter light stream; and separating paraffins from the cooled splitter light stream. An embodiment of the present disclosure is one, any or all of prior embodiments in this paragraph up through the first embodiment in this paragraph further comprising separating the cooled splitter light stream in a splitter receiver into a splitter overhead vapor stream and a splitter overhead liquid stream; fractionating the splitter overhead liquid stream in the splitter column; and taking the paraffinic stream from the splitter overhead vapor stream. An embodiment of the present disclosure is one, any or all of prior embodiments in this paragraph up through the first embodiment in this paragraph further comprising separating the paraffinic stream in a first fractionation column to provide the first stream and the second stream; and fractionating the first compressed stream in a second fractionation column to provide the overhead stream and the bottom stream comprising C2+ hydrocarbons. An embodiment of the present disclosure is one, any or all of prior embodiments in this paragraph up through the first embodiment in this paragraph further comprising cooling the first stream to provide a cooled first stream; separating the cooled first stream to provide a second vapor stream and a second liquid stream; and passing the second vapor stream to the fractionation column. An embodiment of the present disclosure is one, any or all of prior embodiments in this paragraph up through the first embodiment in this paragraph further comprising compressing the second vapor stream to provide a second compressed stream; and passing the second compressed stream to the fractionation column.

[0079] A second embodiment of the present disclosure is a process for producing paraffins comprising contacting a naphtha stream with a catalyst in the presence of hydrogen to produce paraffins; taking a paraffinic stream from the paraffins; separating the paraffinic stream to provide a first stream comprising C3− hydrocarbons and a second stream comprising C3+ hydrocarbons; and fractionating the first stream in a fractionation column to provide an overhead stream and a bottoms stream comprising C2+ hydrocarbons. An embodiment of the present disclosure is one, any or all of prior embodiments in this paragraph up through the second embodiment in this paragraph further comprising fractionating a reactor effluent stream taken from the paraffins in a splitter column to provide a splitter heavy stream comprising aromatics and a splitter light stream comprising paraffins; and taking the paraffinic stream from the splitter light stream.

[0080] A third embodiment of the present disclosure is a process for producing paraffins comprising fractionating a paraffins stream in a splitter column to provide a splitter light stream comprising paraffins and a splitter heavy stream comprising aromatics; separating the splitter light stream to provide a first stream comprising C3− hydrocarbons and a second stream comprising C3+ hydrocarbons; and fractionating the first stream in a fractionation column to provide an overhead stream and a bottom stream comprising C2+ hydrocarbons.

[0081] Without further elaboration, it is believed that using the preceding description that one skilled in the art can utilize the present disclosure to its fullest extent and easily ascertain the essential characteristics of this disclosure, without departing from the spirit and scope thereof, to make various changes and modifications of the disclosure and to adapt it to various usages and conditions. The preceding preferred specific embodiments are, therefore, to be construed as merely illustrative, and not limiting the remainder of the disclosure in any way whatsoever, and that it is intended to cover various modifications and equivalent arrangements included within the scope of the appended claims.

[0082] In the foregoing, all temperatures are set forth in degrees Celsius and, all parts and percentages are by weight, unless otherwise indicated.

Claims

1. A process for separating paraffins comprising: separating a paraffinic stream to provide a first stream comprising C3− hydrocarbons and a second stream comprising C3+ hydrocarbons; compressing said first stream to provide a first compressed stream; and fractionating said first compressed stream in a fractionation column to provide an overhead stream and a bottom stream comprising C2+ hydrocarbons.
2. The process of claim 1 further comprising: cooling said first compressed stream to provide a first cooled stream; separating said first cooled stream to provide a first vapor stream and a first liquid stream; and passing said first vapor stream to the fractionation column.
3. The process of claim 2 further comprising passing a portion of said first c liquid stream to the fractionation column.
4. The process of claim 2 further comprising: passing said second stream to a depropanizer column to provide an overhead stream comprising propane and a depropanized bottom stream comprising C4+ hydrocarbons.
5. The process of claim 4 further comprising: passing said bottom stream comprising C2+ hydrocarbons to a deethanizer column to provide an overhead product stream comprising ethane, and a deethanized bottoms stream comprising propane; and combining said deethanized bottoms stream with said overhead stream comprising propane to provide a propane product stream.
6. The process of claim 1 further comprising: fractionating a reactor effluent stream in a splitter column to separate aromatics in a splitter heavy stream and provide a splitter light stream comprising paraffins; and taking said paraffinic stream from said splitter light stream.
7. The process of claim 6 further comprising: contacting a naphtha stream with a catalyst in the presence of hydrogen to produce paraffins; and taking said reactor effluent stream from said paraffins.
8. The process of claim 7 further comprising recycling said splitter heavy stream to said contacting step.
9. The process of claim 2, wherein cooling said first compressed stream to provide a first cooled stream comprises: heat exchanging said first compressed stream with the bottom stream comprising C2+ hydrocarbons to cool said first compressed stream and heat said bottom stream comprising C2+ hydrocarbons.
10. The process of claim 5 further comprising: taking a first side stream from the fractionation column; and fractionating said first side stream in the deethanizer column.
11. The process of claim 10, wherein cooling said first compressed stream to provide a first cooled stream comprises: heat exchanging the first compressed stream with said first side stream to cool said first compressed stream and heat said first side stream.
12. The process of claim 10, wherein cooling said first compressed stream to provide a first cooled stream comprises: heat exchanging said first compressed stream with said bottom stream comprising C2+ hydrocarbons and said first side stream to cool said first compressed stream and heat said bottom stream comprising C2+ hydrocarbons and said first side stream.
13. The process of claim 6 further comprising: heat exchanging said splitter light stream with a deethanized bottoms stream in a deethanizer reboiler to cool said splitter light stream and provide a cooled splitter light stream; and separating paraffins from said cooled splitter light stream.
14. The process of claim 13, wherein the step of separating paraffins comprises: separating said cooled splitter light stream in a splitter receiver into a splitter overhead vapor stream and a splitter overhead liquid stream; fractionating said splitter overhead liquid stream in the splitter column; and taking said paraffinic stream from said splitter overhead vapor stream.
15. The process of claim 1 further comprising: separating said paraffinic stream in a first fractionation column to provide said first stream and said second stream; and fractionating said first

compressed stream in a second fractionation column to provide said overhead stream and said bottom stream comprising C2+ hydrocarbons.

16. The process of claim 1 further comprising: cooling said first stream to provide a cooled first stream; separating said cooled first stream to provide a second vapor stream and a second liquid stream; and passing said second vapor stream to the fractionation column.

17. The process of claim 16 further comprising: compressing said second vapor stream to provide a second compressed stream; and passing said second compressed stream to the fractionation column.

18. A process for producing paraffins comprising: contacting a naphtha stream with a catalyst in the presence of hydrogen to produce paraffins; taking a paraffinic stream from said paraffins; separating said paraffinic stream to provide a first stream comprising C3− hydrocarbons and a second stream comprising C3+ hydrocarbons; and fractionating said first stream in a fractionation column to provide an overhead stream and a bottoms stream comprising C2+ hydrocarbons.

19. The process of claim 16 further comprising fractionating a reactor effluent stream taken from said paraffins in a splitter column to provide a splitter heavy stream comprising aromatics and a splitter light stream comprising paraffins; and taking said paraffinic stream from said splitter light stream.

20. A process for producing paraffins comprising: fractionating a paraffins stream in a splitter column to provide a splitter light stream comprising paraffins and a splitter heavy stream comprising aromatics; separating said splitter light stream to provide a first stream comprising C3− hydrocarbons and a second stream comprising C3+ hydrocarbons; and fractionating said first stream in a fractionation column to provide an overhead stream and a bottom stream comprising C2+ hydrocarbons.